

# Assignment Specification

IoT Standards and Protocols

# Context

- 100% of your overall mark!
  - Project ethos (1 overall project)
- For this assignment you are required to:
  - Propose a project.
  - Create a working project using networking/IoT standards & protocols.
  - Requires a "physical" aspect – sensors/devices(or simulated by request)
  - Present and communicate your work in a clear, correct manner.
- The project should incorporate the different layers (sensor, processing node, gateway, application) of an IoT/connected device architecture.
- Project will be assessed on its technical (e.g., features) content, complexity, applicability to domain and project communication/execution

# Requirements

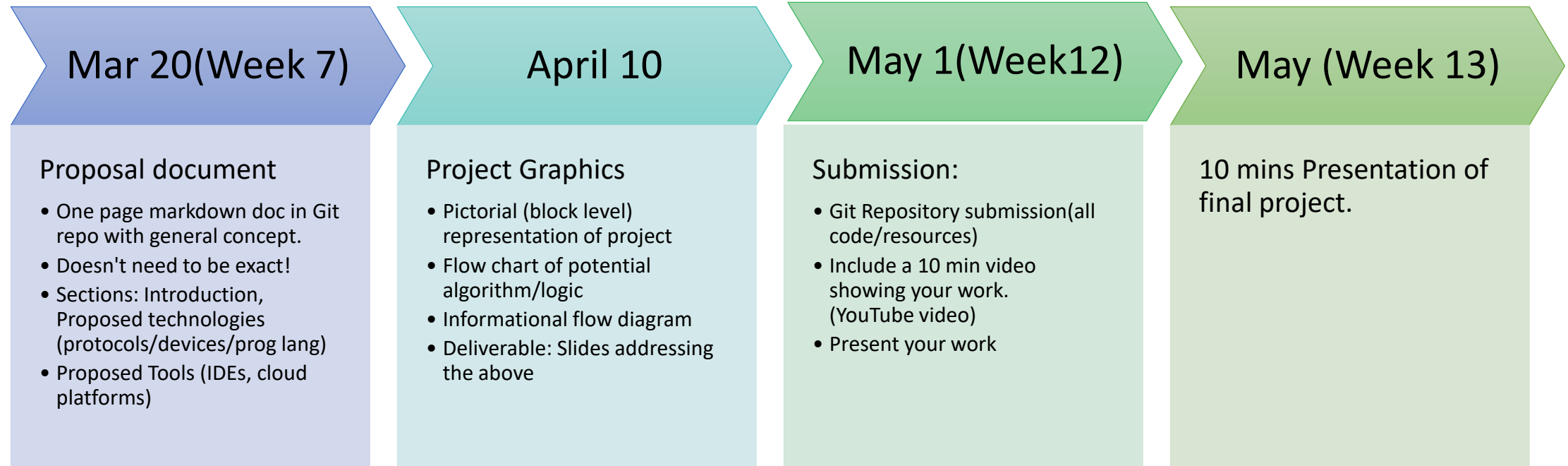
"Include the different layers (sensor, processing, gateway, application) in an IoT Project."

|                     |                                                                                                                                                                                                      |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Design              | Propose and design solutions (e.g. See attached work packages)                                                                                                                                       |
| Apply               | Apply suitable IoT protocols and standards.                                                                                                                                                          |
| Model/<br>implement | Model/implement a solution to your proposal<br>Use the knowledge, skills and practices from other modules<br>Should be scoped correctly – can't build a production standard solution in a few weeks! |
| Present/curate      | Present/curate your project<br>Create a short video that demonstrates the project.                                                                                                                   |

# IoT Layers of your Project

- **Sensor Layer:** Collects real-time data from the environment.
- **Processing Node:** Raspberry Pi processes and transmits data.
- **Gateway Layer:** Data is relayed through an IoT hub/platform.
- **Application Layer:** Visualises and actuates responses.

# Timeline

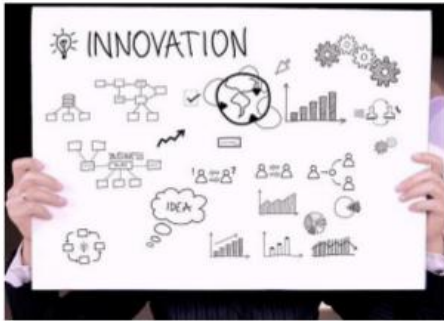


# Assignment Requirements

Each person/team must **propose, design, and implement** an **IoT-based solution** that includes:

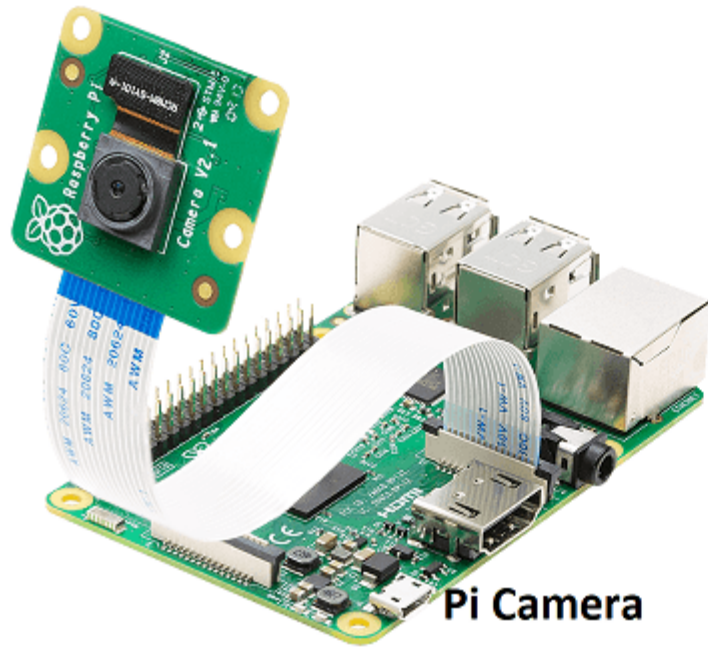
| Requirement                          | Description                                                                                                                                                                                                                                                                                                                     | Weight |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| IoT Device Connection                | - Set up a <b>Raspberry Pi 4 or similar device</b> connected to an <b>IoT messaging platform or server</b> . - Configure <b>MQTT communication or HTTP-based messaging</b> . - Verify <b>data transmission from the device to the platform/server</b> .                                                                         | 10%    |
| Sensor Data Collection               | - Choose <b>at least two sensors</b> for monitoring (e.g., temperature, motion, light, etc.). - Write a <b>Python script</b> to collect <b>real-time data</b> and store it locally. - Ensure the <b>data is timestamped</b> .                                                                                                   | 15%    |
| Telemetry Transmission               | - Format sensor readings as <b>JSON</b> . - Implement <b>secure data transmission using MQTT or HTTP</b> to the IoT server/platform. - Apply <b>error handling and retry mechanisms</b> .                                                                                                                                       | 15%    |
| Data Visualisation                   | - Use a <b>dashboard or visualisation tool</b> (e.g., IoT dashboard/ web interface) to display sensor trends. - Display key parameters (e.g., <b>Heart Rate, motion activity</b> ). - Ensure <b>real-time updates</b> are reflected on the dashboard.                                                                           | 15%    |
| Data Processing & Actuation          | - Implement an <b>automated response based on sensor readings</b> . - Use a <b>server-side script or event-processing system</b> to analyse incoming data and trigger actions (e.g., turning on a fan, sending an alert). - If applicable, integrate <b>computer vision or AI-based analysis</b> for real-time decision-making. | 30%    |
| Project Presentation & Documentation | Submit a <b>report</b> that includes: - <b>System architecture diagrams</b> . - <b>Code snippets and explanations</b> . - <b>Screenshots of dashboards and telemetry data</b> . - <b>Challenges encountered and solutions implemented</b> . - A <b>5-minute demonstration video</b> showcasing the project.                     | 15%    |

# Other Considerations



- "Permissionless Innovation"!
  - You don't have to limit yourself to module technologies. You can integrate other devices/ systems
  - We used Python but you can use any other programming languages/ frameworks if you want. (Remember, the RPi can run Node/Java/JavaScript)
- Have a look at other IoT project examples for inspiration.
- It's OK to "simulate sensors"
  - You can simulate data (e.g. GPS location.)

# Possible Concept: Home Monitoring/Ambient/Assisted Living



# ProjectGraphic

